Exercise 2.12. For each pair of integers, find the unique quotient and remainder when b is divided by a.

(a) $a = 4$, $b = 15$
(b) $a = 15$, $b = 4$
(c) $a = -7$, $b = 16$
(d) $a = 5$, $b = -11$
(e) $a = -1$, $b = -15$
(f) $a = 4$, $b = 0$

Solution.

For division of integers, we use the **division algorithm**:

$$b = aq + r, \qquad 0 \le r < |a|,$$

where q is the **quotient** and r is the **remainder**.

(a) $a = 4,\ b = 15$

We want

$$15 = 4q + r.$$

Since $15 = 4(3) + 3$,

$$\boxed{q = 3, \quad r = 3}.$$

(b) $a = 15,\ b = 4$

$$4 = 15(0) + 4.$$

Thus,

$$\boxed{q = 0, \quad r = 4}.$$

(c) $a = -7,\ b = 16$

We need

$$16 = (-7)q + r.$$

Taking $q = -2$,

$$16 = (-7)(-2) + 2 = 14 + 2.$$

Thus,

$$\boxed{q = -2, \quad r = 2}.$$

Notice that the remainder is positive and satisfies $0 \le 2 < 7$.

(d) $a = 5,\ b = -11$

We need

$$-11 = 5q + r.$$

Taking $q = -3$,

$$-11 = 5(-3) + 4 = -15 + 4.$$

Thus,

$$\boxed{q = -3, \quad r = 4}.$$

(e) $a = -1,\ b = -15$

$$-15 = (-1)(15) + 0.$$

Thus,

$$\boxed{q = 15, \quad r = 0}.$$

(f) $a = 4,\ b = 0$

$$0 = 4(0) + 0.$$

Thus,

$$\boxed{q = 0, \quad r = 0}.$$